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VIDEO PREDICTION POLICY 2: PREDICT BETTER, ACT BETTER

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Project Page: `https://robert-gyj.github.io/video-prediction-policy-2`

ABSTRACT

World action models (WAMs) have emerged as an important class of generalist robot policies, aiming to transfer video prediction priors to action learning. However, we find that existing WAMs frequently produce incorrect motion predictions in open-ended environments, leading to erroneous actions. We attribute this limitation to two factors: (1) base video models are not optimized for manipulation, and (2) naively incorporating action components into video models can substantially degrade their generalization capabilities. We introduce Video Prediction Policy 2 (VPP2), a WAM that enables strong zero-shot generalization in both video prediction and action generation. First, we curate a large-scale, diverse dataset of manipulation videos to continue pretraining the base video foundation model. We annotate video clips with detailed captions and perform *event-level* video pretraining to promote generalization across open-ended manipulation tasks. Second, we post-train and distill the video model into a single-step visual planner with a fixed prediction horizon. Finally, we introduce an action module via a mixture-of-transformers (MoT) architecture to learn an implicit inverse dynamics model. Experiments demonstrate three key results: (1) VPP2-14B outperforms Cosmos3-64B by 11.0% points in video prediction instruction-following success rate on open-ended tasks; (2) VPP2 surpasses the strongest baseline by 18.5% points in success rate on real-world zero-shot ALOHA manipulation tasks; and (3) following benchmark-specific post-training, VPP2 achieves the highest success rates among evaluated methods on the challenging LIBERO-Pro, LIBERO-OOD, and RoboDojo benchmarks.

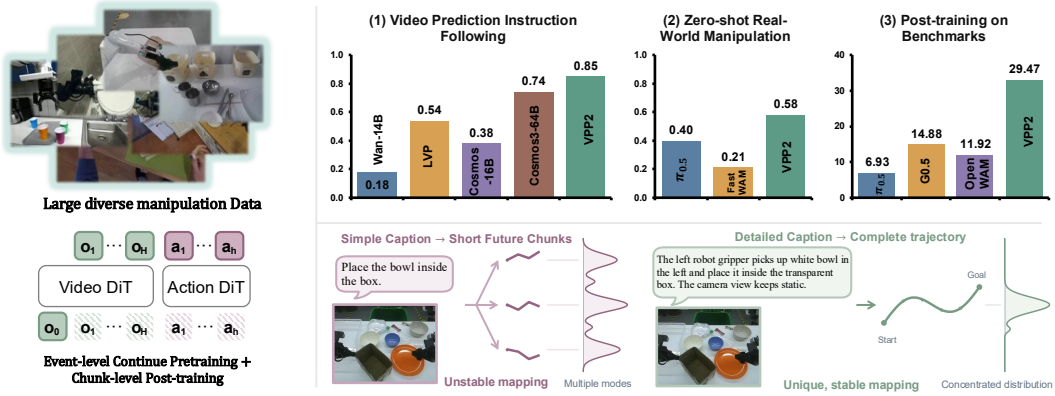


Figure 1: A typical robot policy maps a simple instruction and an observation into a short action chunk. As datasets scale, this mapping becomes increasingly multimodal and uncertain, leading models to learn spurious short-horizon correlations. VPP2 first establishes consistent semantic-to-trajectory mappings via *next event video prediction* with detailed captions, then post-trains and distills the model to generate short action chunks.

1 INTRODUCTION

World action models (WAMs) are developing rapidly and have become an important class of generalist robot policies (Hu et al., 2024; Liao et al., 2025; Kim et al., 2026; Yuan et al., 2026; Ma et al., 2026; Li et al., 2026a;b; AgiBot Research Team et al., 2026; Ye et al., 2026). Many WAMs build on video foundation models (Wan et al., 2025; Agarwal et al., 2025; Yang et al., 2025), aiming to transfer their rich priors about physical dynamics to policy learning. This premise is compelling: accurate, instruction-conditioned video predictions can guide action generation through an explicit or implicit inverse dynamics model (Du et al., 2023; Hu et al., 2024). However, prior work (Zhang et al., 2026b) has found that existing WAMs often perform well only on a narrow set of seen tasks, while producing unreliable future motion predictions in unseen, out-of-distribution scenarios. These prediction failures, in turn, lead to erroneous actions. In recent benchmarks that require generalization, robot policies, including WAMs, with near-perfect success rates on standard evaluation suites suffer substantial performance drops under perturbed task configurations and novel skill compositions (Zhou et al., 2025; Li, 2025; Mishra et al., 2026).

We identify two key factors that limit the generalization of existing WAMs to unseen scenarios. First, base video models are primarily optimized for creative content generation and aesthetic quality rather than precise physical dynamics, and therefore often fail to follow manipulation instructions faithfully (Chen et al., 2025; Zhang et al., 2026b). Second, introducing action-specific components or training objectives into pretrained video foundation models can substantially degrade their generalization capabilities (Mishra et al., 2026).

To address these issues, we introduce Video Prediction Policy 2 (VPP2), a WAM that achieves strong zero-shot generalization in both video prediction and action generation. Our first objective is to train a generalizable video foundation that can faithfully follow instructions and make future predictions grounded in the current observation, without unsupported changes to the scene or task. To this end, we annotate each clip with a detailed caption specifying the manipulation process, active end effector, and target object, and perform *event-level* video prediction training. As illustrated in Figure 1, detailed captions and complete event-level trajectory prediction substantially reduce uncertainty in future prediction and encourage consistent mappings from semantic descriptions to visual trajectories, improving generalization across diverse instructions. We then post-train the video model to predict fixed-horizon future chunks and apply consistency distillation (Song et al., 2023) to obtain a single-step visual planner for real-time robot execution. Finally, we introduce an action expert through a mixture-of-transformers (MoT) architecture (Liang et al., 2025) to learn an implicit inverse dynamics model conditioned on the predicted future. During policy execution, we pair VPP2 with a VLM planner (Bai et al., 2025; Bytedance Seed, 2026) that translates open-ended instructions into explicit subtask instructions.

Our experiments demonstrate three key advantages of VPP2. (1) VPP2-14B follows manipulation instructions faithfully, outperforming Cosmos3-64B (Agarwal et al., 2026) by 11% in instruction-following success rate on open-ended tasks. (2) VPP2 exhibits strong zero-shot manipulation capabilities on a real-world ALOHA platform (Zhao et al., 2023), achieving an average success rate of 58.5% across 10 task categories, compared with 40.0% for $\pi_{0.5}$ (Intelligence et al., 2025) and 20.5% for Fast-WAM (Yuan et al., 2026). (3) Following benchmark-specific post-training, VPP2 achieves the highest success rates among evaluated methods on LIBERO-Pro (Zhou et al., 2025) (45.0%, compared with 11.0% for the strongest baseline), LIBERO-OOD (Li, 2025; Mishra et al., 2026) (63.9%), and RoboDojo (Chen et al., 2026) (32.26%, state-of-the-art).

2 DATA PROCESSING PIPELINE

In this section, we describe our data collection and curation pipeline, including how we unify the inputs and outputs across diverse manipulation datasets.

Video Sources. We compile a large-scale, diverse collection of manipulation videos for continued pretraining of an open-source video foundation model (Wan et al., 2025). Our data span three categories: robot manipulation datasets, human activity datasets, and general-purpose video datasets. For general-purpose video datasets, we apply an extensive filtering pipeline similar to that of LVP (Chen et al., 2025) to retain only videos depicting manipulation-related activities. Table 1 summarizes the datasets used for training; full dataset references are provided in Appendix A.

Data Type	Embodiment Type	Data Sources
Robot	Single-arm	OXE, RoboMIND, DROID, RH20T, MolmoAct
	Dual-arm	AgiBot World-Beta, RoboCOIN, RDT, GM-100, ABC-130K, Self-collected ALOHA Data
	Mobile & humanoid	InternData-A1, Galaxea Open-World, Self-collected dexterous hands
Human	Human hands	Open-source: EgoDex, Ego4D, EPIC-KITCHENS, Something-Something V2, VITRA-1M Self-collected egocentric data
General Video	Human hands	OpenVid, Panda-70M

Table 1: Overview of training data sources. Full dataset references are provided in Appendix A, Table 5.

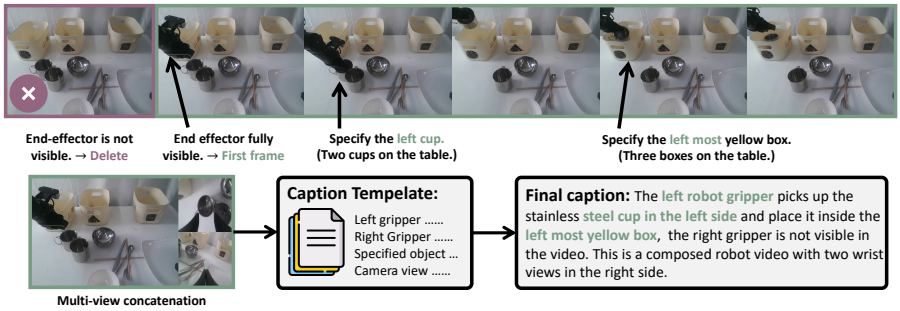
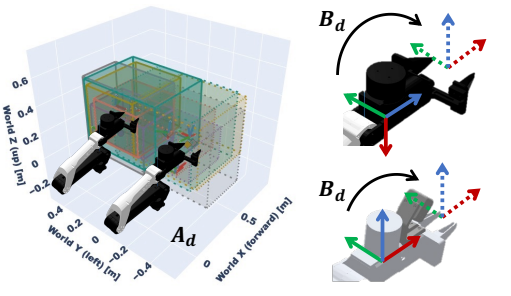


Figure 2: An example of our data processing pipeline. The central objective is to reduce uncertainty in future prediction, thereby encouraging the model to learn a consistent mapping from conditioning information to sub-task trajectories. To this end, each caption includes detailed task descriptions, explicit end-effector identification, a description of the visible target object, and camera-view changes.

Video Filtering, Segmentation and Captioning. We first remove corrupted trajectories, including those with camera failures or abrupt discontinuities, and segment the remaining trajectories into semantically coherent clips lasting 1–12 seconds. We identify candidate boundaries using heuristic cues associated with natural transitions, such as local minima in human-hand or end-effector velocity and moments when the gripper or fingers open or close. A vision-language model (VLM) then refines the segmentation by merging adjacent clips where appropriate and adjusting their start and end points.

After segmentation, we generate detailed captions that describe the task, explicitly identify the active end effectors, unambiguously specify the target objects, and characterize the camera views and motion. Figure 2 illustrates an example, and Appendix A.1 provides further details.

Unified Action Space and Unified End-effector Coordinate Systems. Different robot datasets often adopt inconsistent coordinate systems and camera viewpoints, resulting in conflicting action representations for visually similar motions. For example, a leftward motion may correspond to the positive x-direction in one dataset but the negative y-direction in another, making it difficult for the model to learn the semantic meaning of each action dimension.



(a) Workspace alignment (b) End-effector alignment

Figure 3: (a) Bounding boxes show the aligned workspaces of different datasets. (b) Aligned end-effector coordinate frames.

We focus on egocentric bimanual datasets, including ALOHA-style, humanoid-style, and human egocentric datasets. These datasets account for more than 80% of the total data and share a similar bimanual structure and camera viewpoint. We explicitly align their coordinate systems at two levels: (1) *workspace alignment*, which applies a world-frame transformation so that different robots have comparable end-effector workspaces; and (2) *end-effector alignment*, which applies a local transformation to standardize end-effector origins and axis orientations.

For an arm dataset d , let $T_d = {}^{W_d}T_{E_d} \in SE(3)$ denote the original end-effector pose, where W_d and E_d are the dataset's world and end-effector frames, respectively. We define the workspace alignment as $A_d = {}^{\bar{W}}T_{W_d}$ and the end-effector alignment as $B_d = {}^{E_d}T_{\bar{E}}$, where $\bar{W}$ and $\bar{E}$ denote the canonical world and end-effector frames. The aligned end-effector pose is given by $\widetilde{T}_d = A_d T_d B_d$,.

Unified T-shape Multi-view Input across Datasets. Robotics datasets commonly contain observations from multiple camera views. We arrange them into a T-shaped composite image, with the primary view on the left and two auxiliary views stacked vertically on the right, as illustrated in the bottom-left panel of Figure 2. When fewer than two auxiliary views are available, we fill the missing slots with black placeholder images.

3 VPP2: A Generalist Policy with Zero-shot Capability

Overview. Our large-scale, diverse, and densely annotated dataset enables us to train a policy with strong generalization capabilities. VPP2 builds upon the pretrained Wan2.1-I2V-14B model (Wan et al., 2025) and introduces an action expert via a mixture-of-transformers (MoT) architecture (Liang et al., 2025). Our training objective is twofold: we first adapt the video model to produce generalizable future predictions at real-time inference speed for open-ended manipulation tasks, and then train the action expert conditioned on the KV cache of the video model. To maximize policy generalization, we organize training into the stages summarized in Figure 4.

3.1 Video Prediction Model Training Pipeline

Stage 1: Event-level Video Model Continued Pre-training. We first adapt a pretrained video generation model to the manipulation domain through event-level continued pre-training. Each training example covers a complete manipulation subtask, aligning the subtask description with the corresponding visual evolution from the initial observation to subtask completion.

Given a demonstration $\tau = (o_0, o_1, \ldots, o_T)$, we uniformly sample N future frames across the entire event:

$$\mathbf{y}_{\text{event}} = \left(o_0, o_{\lfloor T/N \rfloor}, o_{\lfloor 2T/N \rfloor}, \ldots, o_T\right), \tag{1}$$

where T is the final frame index and N excludes the initial conditioning frame o_0. Let $\mathbf{x}_1 = \mathcal{E}(\mathbf{y}_{\text{event}})$ denote the corresponding video latent representation, where $\mathcal{E}$ is the video encoder. We construct the interpolated latent as

$$\mathbf{x}_s = (1-s)\boldsymbol{\epsilon} + s\mathbf{x}_1, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad s \in [0,1], \tag{2}$$

where s denotes flow time, with $s = 0$ corresponding to noise and $s = 1$ to data. The model is optimized with the flow matching objective (Lipman et al., 2023)

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{(\mathbf{x}_1, c),\, \boldsymbol{\epsilon},\, s}\left[\left\| v_\theta(\mathbf{x}_s, s, c) - (\mathbf{x}_1 - \boldsymbol{\epsilon})\right\|_2^2\right], \tag{3}$$

where v_θ predicts the flow velocity and c includes the subtask instruction and initial observation o_0.

In this stage, we predict 49 frames at a resolution of 416×240 and continue pre-training Wan2.1-I2V-14B for $30{,}000$ optimization steps with batch size $1{,}024$ and learning rate 1e-5. Our experiments in Sec. 4.1 show that the resulting model achieves strong instruction following on manipulation tasks, outperforming the 64B Cosmos3-Super-Image2Video model (Agarwal et al., 2026).

Stage 2: Chunk-level Video Model Post-training. Event-level continued pre-training enables the model to generalize well to manipulation tasks. However, sampling a fixed number of frames from events of different durations produces variable temporal intervals between predicted frames, complicating their alignment with downstream actions. We therefore further post-train the model to predict a fixed-duration future chunk, establishing a consistent temporal scale for action learning. In experiments, we predict 2 seconds for human-hand manipulation and 8 seconds for robot manipulation.

Let H denote the prediction horizon in seconds and f the dataset frame rate in frames per second (fps). The nominal interval between sampled frames is $\delta = fH/N$, measured in dataset frames. For a chunk starting at frame t, the target video is

$$\mathbf{y}^{(t)}_{\text{chunk}} = \left(o_t, o_{t+\lfloor \delta \rfloor}, o_{t+\lfloor 2\delta \rfloor}, \ldots, o_{t+\lfloor N\delta \rfloor}\right), \tag{4}$$

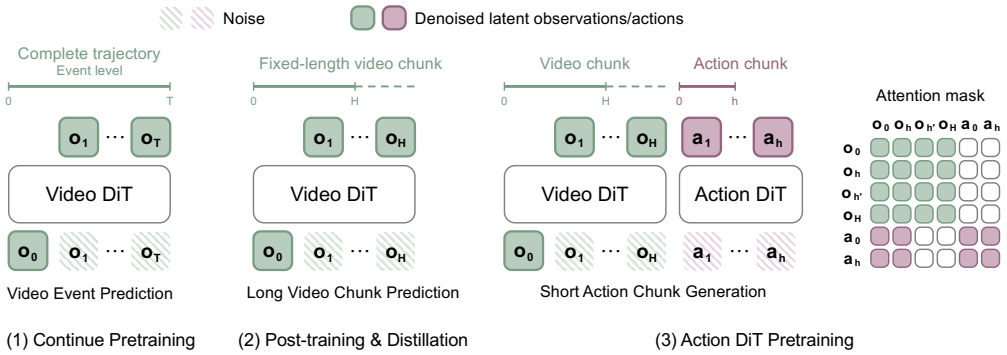


Figure 4: The VPP2 training pipeline is designed to maximize generalization. Stage 1 uses large-scale, event-level video pretraining to learn a generalizable video model for manipulation. Stage 2 post-trains and distills this model to predict long-horizon video chunks spanning 8 seconds in a single forward pass, taking approximately 0.1 seconds. Finally, Stage 3 trains an action expert to generate 2-second action chunks conditioned on the one-step video latents.

with conditioning inputs c containing the observation o_t and the corresponding subtask instruction. The index $t + \lfloor fH \rfloor$ is capped at the final frame index T, and we optimize the same flow matching objective in Eq. equation 3, with $\mathbf{x}_1 = \mathcal{E}(\mathbf{y}^{(t)}_{\text{chunk}})$.

Stage 2: Consistency Distillation. We distill the chunk-level video model for single-step generation (Song et al., 2023) by enforcing consistent terminal predictions at adjacent flow times:

$$\mathcal{L}_{\text{CD}} = \mathbb{E}\left[\lambda(s, s') \left\| F_\theta(\mathbf{x}_s, s, c) - \text{stopgrad}\left(F_{\bar{\theta}}(\widehat{\mathbf{x}}_{s'}, s', c)\right)\right\|_2^2\right], \tag{5}$$

where $0 \le s < s' \le 1$, $\widehat{\mathbf{x}}_{s'}$ is obtained by integrating the frozen teacher flow from $(\mathbf{x}_s, s)$ to s', and $\bar{\theta}$ denotes the EMA student parameters. The student satisfies $F_\theta(\mathbf{x}, 1, c) = \mathbf{x}$. To emphasize single-step generation, we sample $s' = 1$ with probability 0.5; otherwise, we sample s' from the remaining flow times. At inference, the video latent is generated in a single forward pass as $F_\theta(\boldsymbol{\epsilon}, 0, c)$.

3.2 Action Modeling

Stage 3: Action Pretraining across Datasets. After training a generalizable video model, we train the action expert on our processed bimanual datasets with unified workspace and end-effector coordinate systems. The action expert is a 0.9B-parameter diffusion transformer (DiT) (Peebles & Xie, 2023) with a standard MoT architecture. Since the action expert is newly initialized, we freeze the base parameters of the video DiT during the early stages of action training and adapt the video backbone using only LoRA (Hu et al., 2022). This strategy helps preserve the pretrained video representations while limiting disruption from action-training gradients.

Inference Latency. We generate video efficiently by reducing the video input to $17 \times 416 \times 240$ during post-training and distilling the video model for one-step generation, as described in Sec. 3.1. With bfloat16 precision and torch.compile, the video model takes approximately 0.12 seconds. The action expert then generates an action chunk conditioned on the video latents using five denoising steps, taking approximately 0.1 seconds. The total latency per action chunk is therefore approximately 0.22 seconds.

3.3 VLM for High-level Planning.

Sub-task Planning. To tackle long-horizon tasks and handle ambiguous human instructions, we adopt a hierarchical architecture in which a VLM generates subtask plans (Shi et al., 2025). The VLM handles high-level reasoning, semantic understanding, and memory, allowing VPP2 to focus on mapping explicit instructions to trajectories. In practice, the VLM can be deployed locally or accessed through an API to a frontier model.

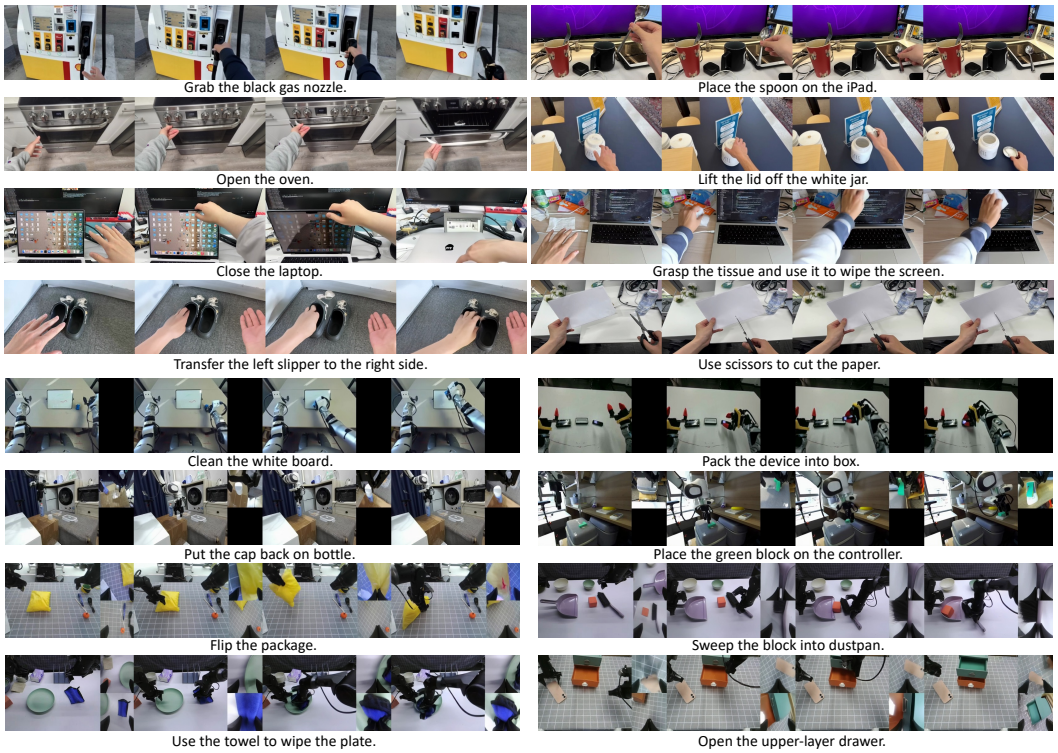


Figure 5: VPP2 video predictions for human-hand and robot manipulation. Trained on large-scale, diverse manipulation datasets, VPP2 generalizes across human hands and a wide range of robot embodiments. For robot manipulation, VPP2 jointly predicts one to three camera views arranged in a T-shaped composite. Due to space constraints, single-step video predictions after distillation are shown in Figure 10 in the appendix.

Prompt Enhancement. VLM planning also generates detailed subtask descriptions that match the caption template used during training. Prior work on video generation likewise highlights the importance of detailed captions for generation quality (Wan et al., 2025; Yang et al., 2025).

4 Experiments

In this section, we conduct experiments to answer the following questions: (1) How does the motion quality of VPP2's video predictions compare with that of state-of-the-art video foundation models on manipulation tasks? (2) How broad are VPP2's zero-shot manipulation capabilities? (3) How well does VPP2 perform after domain-specific post-training?

4.1 Video Prediction Quality Analysis

Video Prediction for Human and Robot Manipulation. Figure 5 visualizes predictions for human-hand manipulation and diverse robot embodiments from the pretrained VPP2 model. To assess generalization, we randomly sample test cases from prior work (Chen et al., 2025; Zhang et al., 2026b) and take photographs in office environments and robot workspaces, pairing them with freely chosen manipulation instructions. Trained on data spanning more than 20 robot embodiments, VPP2 also generalizes to unseen embodiments and tasks, as shown in Figure 5. VPP2 also supports joint prediction across one to three camera views, with the primary view placed on the left and up to two auxiliary views stacked on the right.

Quantitative Comparisons. We compare VPP2 with four video foundation models and one ablation variant: (1) Wan-2.1-I2V-14B-480p (Wan et al., 2025), our base model; (2) LVP (Chen et al., 2025), which also adapts Wan-2.1-I2V-14B-480p to manipulation tasks; (3) Cosmos3-Nano-16B, which is extensively trained on robot manipulation data; (4) Cosmos3-Super-Image2Video-64B, the strongest

Method	Human	Robot
Wan2.1-14B	0.32	0.04
LVP	0.62	0.46
Cosmos3-16B	0.44	0.32
Cosmos3-64B	0.70	0.78
VPP2-Fixed-Step	0.48	0.42
VPP2 (ours)	**0.80**	**0.90**

Table 2: Instruction following success rates.

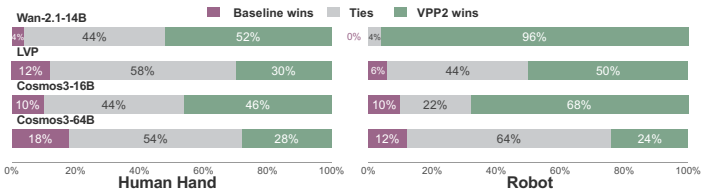
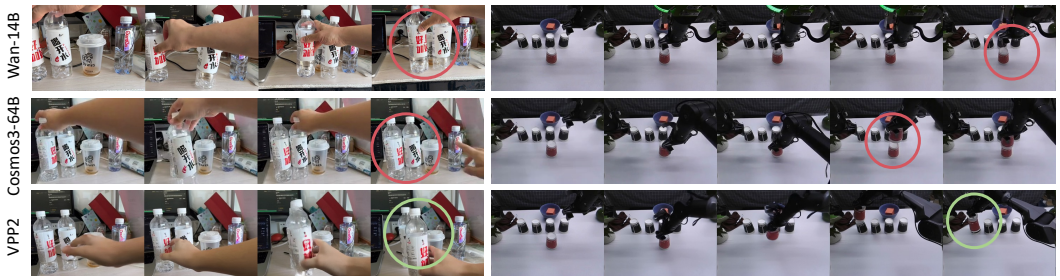


Figure 6: Win rates of the pretrained VPP2 model against different baselines on video prediction tasks.



(a) Grasp leftmost bottle and place it in front of the short cup. **(b) Stack red paper cup on leftmost black paper cup.**

Figure 7: Comparison of instruction-following capabilities among VPP2, Wan-14B, and Cosmos3-64B. VPP2 demonstrates better instruction following on complex tasks requiring spatial understanding.

model in the Cosmos 3 family (Agarwal et al., 2026); and (5) VPP2-Fixed-Step, an ablation of our method that predicts a fixed-duration future chunk rather than a complete event, potentially misaligning the predicted future with the instruction describing the full event.

We randomly collect 50 image–instruction pairs each for human-hand and robot manipulation and generate video predictions for every pair using each model. We first use a GPT-based evaluator to assess instruction-following success rates, reported in Table 2. Human evaluators then compare pairs of predictions to determine which better follows the instruction, yielding VPP2's pairwise win rates against each baseline in Figure 6. VPP2's advantages are most pronounced in complex manipulation scenarios requiring spatial understanding, as illustrated in Figure 7. We hypothesize that training with detailed captions encourages more accurate mappings from instructions to trajectories, contributing to these improvements.

4.2 Policy Performance Analysis

Zero-shot Performance on Real-world Aloha Robot. After pretraining a strong video foundation model for manipulation, we post-train and distill it into a fast, single-step generator to support action learning. With its generalizable video predictions, VPP2 demonstrates strong zero-shot capabilities across open-ended manipulation tasks. We deploy VPP2 directly on a robot embodiment seen during training, without task-specific fine-tuning. For comparison, we train $\pi_{0.5}$ (Intelligence et al., 2025) and the Wan-14B version of Fast-WAM (Yuan et al., 2026) on the same data used in VPP2 training. We evaluate models on 10 different categories of zero-shot tasks and present results in Figure 8. VPP2 achieves an average success rate of 58.5%, compared with 40.0% for $\pi_{0.5}$ and 20.5% for Fast-WAM, and performs best in 9 of the 10 categories.

LIBERO-ID, LIBERO-Pro, and LIBERO-OOD Benchmarks. All models are trained exclusively on the four standard LIBERO suites (Liu et al., 2023) and are evaluated under three complementary settings. LIBERO-ID measures standard in-distribution manipulation performance. To assess generalization beyond the training distribution, we further evaluate on LIBERO-Pro (Zhou et al., 2025) and LIBERO-OOD (Li, 2025) without additional training. Following HarnessVLA (Zhang et al., 2026d), we evaluate the Position and Task perturbations of LIBERO-Pro, which test generalization to changes in object positions and task specifications, respectively. We further follow (Mishra et al., 2026) to evaluate compositional generalization on LIBERO-OOD, which recombines familiar objects,

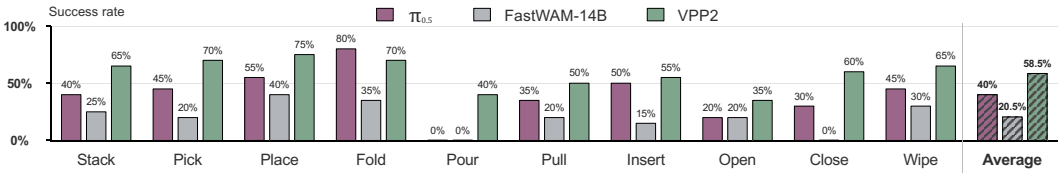


Figure 8: Zero-shot success rates on the real-world ALOHA across 10 randomly selected task categories. For a fair comparison, we finetune $\pi_{0.5}$ and FastWAM on all ALOHA datasets exposed in VPP2's training data and use the 14B variant of FastWAM.

	LIBERO-ID	LIBERO-Pro			LIBERO-OOD			
Method	**Overall**	**Position**	**Task**	**Overall**	**Spatial**	**Object**	**Goal**	**Overall**
π_0 (Black et al., 2024)	94.2	0.5	0.0	0.3	0.7	0.3	4.3	1.7
$\pi_{0.5}$ (Intelligence et al., 2025)	96.9	20.8	1.3	11.0	36.7	2.3	41.7	26.8
MolmoAct (Lee et al., 2025)	86.6	1.5	1.5	1.5	–	–	–	–
X-VLA (Zheng et al., 2026)	98.1	0.8	6.8	3.8	–	–	–	–
AtomVLA (Sun et al., 2026)	97.0	7.3	5.3	6.3	–	–	–	–
Cosmos-Policy (Kim et al., 2026)	98.5	–	–	–	30.7	0.3	0.7	10.5
Fast-WAM (Yuan et al., 2026)	97.6	–	–	–	12.7	0.0	15.7	9.4
DiT4DiT (Ma et al., 2026)	98.6	–	–	–	9.0	0.0	10.3	6.4
Temporal Ratio (Mishra et al., 2026)	94.0	–	–	–	58.6	40.0	80.3	59.4
VPP2 (Ours)	**98.8**	**42.3**	**47.8**	**45.0**	**59.3**	**44.5**	**87.7**	**63.9**

Table 3: Post-training on LIBERO-ID, LIBERO-Pro, and LIBERO-OOD benchmarks. All models are trained exclusively on the four standard LIBERO suites (Liu et al., 2023) and are evaluated under three complementary settings.

layouts, and goals into unseen task configurations along spatial, object, and goal dimensions. Per-suite LIBERO-ID results are provided in Appendix C.1.

As shown in Table 3, while most baselines exceed 94% on LIBERO-ID, this in-distribution performance does not transfer to generalization. On LIBERO-Pro, VLA baselines drop to at most 11.0% overall, whereas VPP2 reaches 45.0%, with the largest gain on the Task perturbation (47.8% vs. 6.8%), indicating that VPP2 follows the given instruction rather than replaying memorized trajectories. On LIBERO-OOD, video-based policies such as Cosmos-Policy, Fast-WAM, and DiT4DiT reach at most 10.5%, while VPP2 achieves 63.9%, also surpassing Temporal Ratio (Mishra et al., 2026), which specifically targets this generalization gap. We attribute these gains to next-event video prediction with detailed captions, which establishes consistent mappings from instructions to future trajectories.

RoboDojo Benchmark. RoboDojo (Chen et al., 2026) is a unified sim-and-real benchmark for evaluating generalist robot manipulation policies. Its simulation benchmark contains 42 bimanual tasks covering five capability dimensions: generalization, memory, precision, long-horizon execution, and open-vocabulary instruction following. We follow the official training and evaluation protocol and report both the success rate and the average score, which captures partial task progress.

As shown in Table 4, VPP2 achieves an average score of 39.26 and a success rate of 32.26%, achieving state-of-the-art performance.[1] It outperforms representative VLAs such as $\pi_{0.5}$ (Intelligence et al., 2025), X-VLA (Zheng et al., 2026), and Xiaomi-Robotics-1 (Guo et al., 2026), world action models such as Fast-WAM (Yuan et al., 2026) and OpenWAM-α (Wang et al., 2026a), and the frontier foundation model GPT-6-Astra (Zhang et al., 2026c) (28.97 / 22.48%). Per-dimension results are provided in Appendix C.2.

VPP2 with Subtask Planning. The VPP2 results in Table 4 use full-task instructions without subtask planning. We further investigate VLM-based subtask planning on five selected RoboDojo task groups covering object classification, block stacking, block swapping, mahjong, and tic-tac-toe. To train the subtask-conditioned policy, we segment demonstrations and pair each segment with a subtask instruction. At test time, a vision-language model (VLM) uses visual observations and execution feedback to select the next subtask for VPP2 to execute.

[1] Baseline results are taken from the official RoboDojo simulation leaderboard as of September 2026.

	π_0	$\pi_{0.5}$	X-VLA	Fast-WAM	OpenWAM-α	Xiaomi-Robotics-1	GPT-6-Astra	**VPP2 (Ours)**
Avg. Score	3.48	11.44	10.13	3.48	17.18	20.07	28.97	**39.26**
Success Rate (%)	1.53	6.93	6.52	2.03	11.92	13.93	22.48	**32.26**

Table 4: Post-training results on the RoboDojo simulation benchmark. Average score captures partial task progress, and success rate measures binary task completion; both are averaged over the five capability dimensions. Baseline results are taken from the official leaderboard.

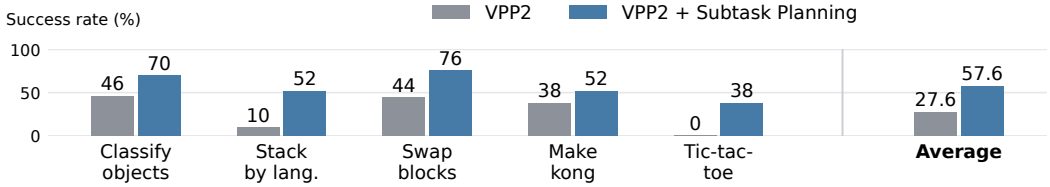


Figure 9: Success rates with and without VLM-based subtask planning (50 trials per task). Average is the five-task mean; the baseline checkpoint matches Table 4.

Figure 9 summarizes the results across the five selected task groups. The evaluated planning configurations achieve an average success rate of 57.6%, compared with 27.6% for the baseline. The largest gains occur in language-conditioned block stacking (10% to 52%) and tic-tac-toe (0% to 38%).

5 Related Works

Video Foundation Model. Video generation has advanced rapidly through latent diffusion models (Blattmann et al., 2023b;a) and large-scale diffusion transformer models (Yang et al., 2025; Wan et al., 2025; Agarwal et al., 2025; 2026). Large-scale bidirectional video models provide strong general-purpose video priors; even recent autoregressive generators build on pretrained bidirectional models, using them for initialization or as teachers before causal adaptation and distillation (Yin et al., 2025; Huang et al., 2025). This motivates our strategy of first learning a generalizable video model and then adapting it for efficient prediction. Besides the training strategy, prompt enhancement is an important component of video foundation models (Yang et al., 2025; Wan et al., 2025). Similarly, we use a VLM to produce detailed, manipulation-specific descriptions that reduce ambiguity in these mappings. After establishing this manipulation-focused video prior, we post-train the model for fixed-horizon prediction and distill it for single-step generation to support efficient action learning.

World Action Model. World action models (WAMs) leverage video prediction to facilitate robot policy learning. Early approaches first generate future frames and then infer actions through inverse dynamics (Du et al., 2023; Black et al., 2023; Bharadhwaj et al., 2024; Liang et al., 2024; Feng et al., 2025), but iterative video generation can incur substantial latency for closed-loop control. Hierarchical approaches also combine video-based motion planning with reactive VLA control (Zhang et al., 2026e). More recent approaches condition actions on features from video models (Hu et al., 2024; Liao et al., 2025; Jia et al., 2025; Yan et al., 2026b) or jointly learn visual prediction and action generation (Zhang et al., 2025; Guo et al., 2024; Li et al., 2025b; Zhu et al., 2025; Ma et al., 2026; Kim et al., 2026; Yan et al., 2026a; Ye et al., 2026; Li et al., 2026a; Bi et al., 2026; Motubrain Team et al., 2026). Recent work also explores event-grounded world-action learning (Li et al., 2026b) and large-scale manipulation-specific pretraining (AgiBot Research Team et al., 2026). Nevertheless, action learning can compromise pretrained capabilities, including video-model generalization (Mishra et al., 2026) and VLM semantic understanding (Zhang et al., 2026a). VPP2 emphasizes instruction-aligned future prediction, combining event-level pretraining with detailed captions, fixed-horizon post-training, and single-step distillation to provide an efficient video backbone for action learning.

6 Conclusion

We presented Video Prediction Policy 2 (VPP2), a world action model that achieves strong zero-shot generalization in both video prediction and action generation. We first continue pretraining the base video model to produce generalizable, instruction-following predictions, then learn a policy while preserving these predictive capabilities. Experiments show that VPP2 follows manipulation

instructions more faithfully than substantially larger video foundation models, exhibits strong zero-shot manipulation capabilities on a real-world ALOHA platform, and achieves the highest success rates after post-training on challenging benchmarks. We hope these findings encourage future WAM research to look beyond architectural choices for action modeling and place greater emphasis on learning generalizable video predictions and preserving this generalization during action learning.

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AI USE STATEMENT

We used generative AI tools to assist with language polishing, improve the clarity and readability of the manuscript, and draft portions of the paper. In the research process, these tools were used to suggest experimental parameter settings and assist with searches for relevant literature and technical information. We also used AI tools to annotate human and robotic manipulation datasets at scale for model training. The authors evaluated the suggestions and made the final decisions regarding experimental design and parameter selection. All AI-assisted text was reviewed and revised by the authors, and retrieved information was checked against original sources before use. The authors take full responsibility for the final content of the paper, including its data, claims, results, and references.

REPRODUCIBILITY STATEMENT

We describe the model architecture, training objectives, and evaluation protocols in the main text, with additional data processing details provided in the appendix. Detailed training configurations, code, and model checkpoints are available on our project website.

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A Dataset Process Details

Dataset Sources. Table 5 provides the full dataset references for the training data summarized in Table 1.

Data Type	Embodiment Type	Data Sources
Robot	Single-arm	OXE (Open X-Embodiment Collaboration, 2023), RoboMIND (Wu et al., 2025a), DROID (Khazatsky et al., 2024), RH20T (Fang et al., 2024), MolmoAct (Lee et al., 2025)
	Dual-arm	AgiBot World-Beta (Bu et al., 2025), RoboCOIN (Wu et al., 2025b), RDT (Liu et al., 2024), GM-100 (Wang et al., 2026b), ABC-130K (Allshire et al., 2026), Self-collected ALOHA Data
	Mobile & humanoid	InternData-A1 (Tian et al., 2025), Galaxea Open-World (Jiang et al., 2025), Self-collected dexterous hands
Human	Human hands	Open-source: EgoDex (Hoque et al., 2025), Ego4D (Grauman et al., 2022), EPIC-KITCHENS (Damen et al., 2018), Something-Something V2 (Goyal et al., 2017; Mahdisoltani et al., 2018), VITRA-1M (Li et al., 2025a) Self-collected egocentric data
General Video	Human hands	OpenVid (Nan et al., 2025), Panda-70M (Chen et al., 2024)

Table 5: Training data sources with references to the original datasets. Self-collected datasets are collected as part of this work.

A.1 Video Captioning.

After segmentation, we use a VLM to generate detailed captions for each clip according to a predefined template. Our captioning pipeline aims to reduce uncertainty in future prediction and encourage consistent mappings from semantic descriptions to visual trajectories. Specifically, we address four major sources of uncertainty:

(1) Detailed Task Description. We describe how the end effector completes the task, including the sequence and manner of manipulation.

(2) Explicit End-Effector Identification. The active end effector must be visible in the first frame; otherwise, we trim the clip to begin when it first becomes visible. When multiple end effectors are involved, we describe the motion of each one explicitly.

(3) Unambiguous Target-Object Specification. When multiple identical or similar objects are present, we identify the target using distinctive attributes or spatial relationships. The target object must also be visible in the first frame.

(4) Camera-View Description. We use optical flow to remove egocentric clips with excessive viewpoint changes. For clips with moderate camera motion, a VLM describes the viewpoint changes and the camera views included in the video, and we incorporate this information into the caption.

A.2 Unified Action Space

We focus on egocentric bimanual datasets, including ALOHA-style, humanoid-style, and human egocentric datasets. These datasets account for more than 80% of the total data and share a similar bimanual structure and camera viewpoint. We explicitly align their coordinate systems at two levels: (1) *workspace alignment*, which applies a world-frame transformation so that different robots have comparable end-effector workspaces; and (2) *end-effector alignment*, which applies a local transformation to standardize end-effector origins and axis orientations.

For an arm dataset d , let $T_d = {}^{W_d}T_{E_d} \in SE(3)$ denote the original end-effector pose, where W_d and E_d are the dataset's world and end-effector frames, respectively. We define the workspace alignment as $A_d = {}^{\bar{W}}T_{W_d}$ and the end-effector alignment as $B_d = {}^{E_d}T_{\bar{E}}$, where $\bar{W}$ and $\bar{E}$ denote the canonical world and end-effector frames. The aligned end-effector pose is given by

$$\widetilde{T}_d = A_d T_d B_d, \tag{6}$$

B More Video Prediction Results

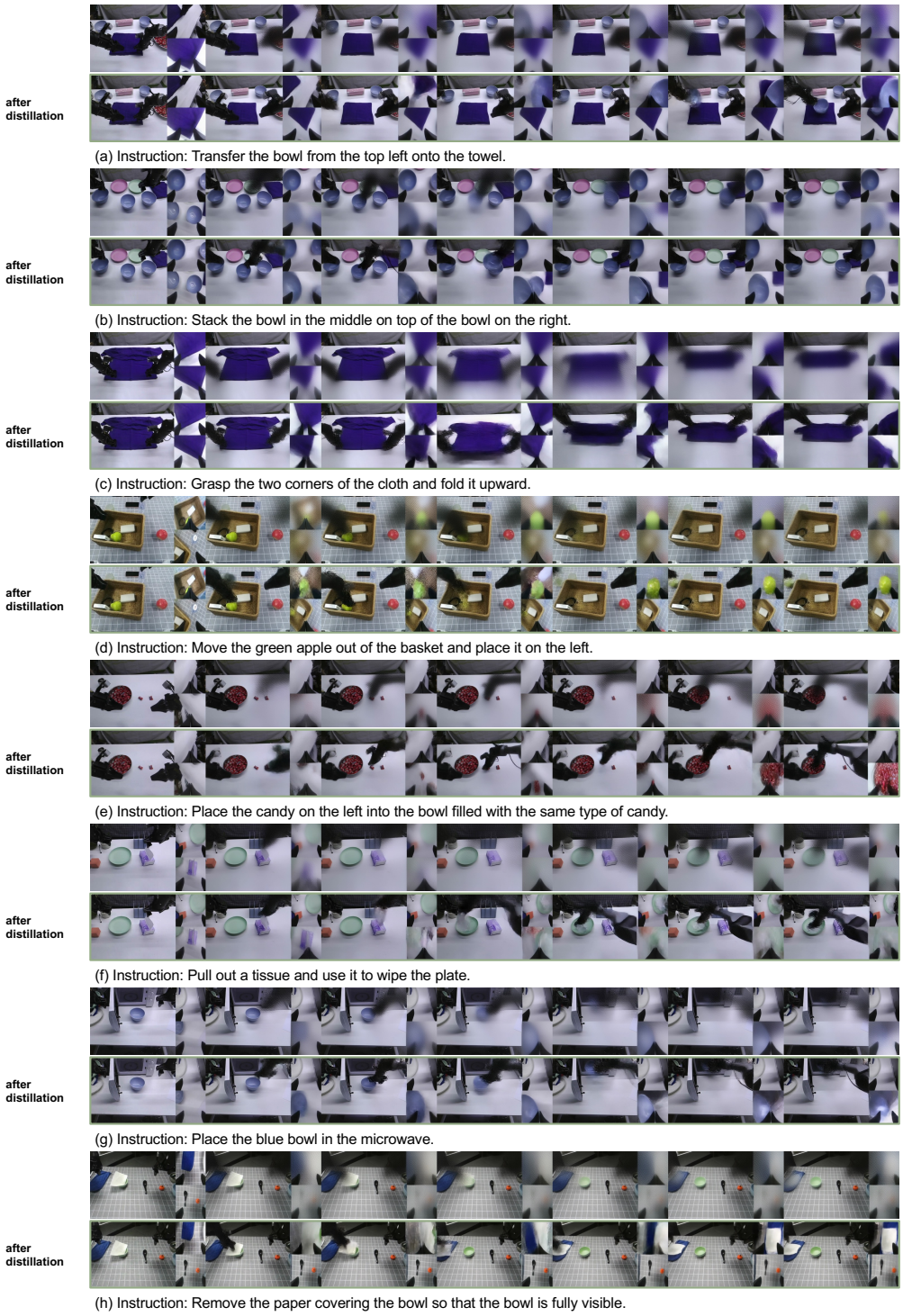


Figure 10: Additional video prediction results on open-ended tasks. We compare **single-step** video predictions before and after distillation. Distillation enables high-quality video prediction with just one sampling step.

C Detailed Benchmark Results

C.1 Detailed LIBERO Results

Table 6 reports per-suite success rates on the four standard LIBERO suites (Liu et al., 2023), which correspond to the LIBERO-ID results in Table 3. Baseline results are taken from the original papers and from Zhang et al. (2026d) and Mishra et al. (2026).

Method	Spatial	Object	Goal	Long	Avg.
π_0 (Black et al., 2024)	96.8	98.8	95.8	85.2	94.2
$\pi_{0.5}$ (Intelligence et al., 2025)	98.8	98.2	98.0	92.4	96.9
MolmoAct (Lee et al., 2025)	87.0	95.4	87.6	77.2	86.6
X-VLA (Zheng et al., 2026)	98.2	98.6	97.8	97.6	98.1
AtomVLA (Sun et al., 2026)	96.4	99.6	97.6	94.4	97.0
Cosmos-Policy (Kim et al., 2026)	98.1	**100.0**	98.2	97.6	98.5
Fast-WAM (Yuan et al., 2026)	98.2	**100.0**	97.0	95.2	97.6
DiT4DiT (Ma et al., 2026)	98.4	99.6	**98.6**	97.6	98.6
Temporal Ratio (Mishra et al., 2026)	96.3	99.6	97.6	82.6	94.0
VPP2 (Ours)	**99.2**	99.8	98.2	**98.0**	**98.8**

Table 6: Per-suite success rates (%) on the standard LIBERO benchmark. Best results in each column are in bold.

C.2 Detailed RoboDojo Results

RoboDojo (Chen et al., 2026) evaluates generalist manipulation policies on 42 bimanual simulation tasks organized into five capability dimensions. *Generalization* tests robustness to unseen backgrounds, lighting, clutter, and target objects, and is evaluated under both standard and randomized settings. *Precision* requires fine-grained target localization and contact-rich control. *Long-Horizon* requires completing all sub-steps of multi-step tasks. *Memory* contains tasks whose correct actions depend on information observed earlier in the episode. *Open* evaluates unseen task specifications whose required skills appear in the training data under different contexts. We follow the official training and evaluation protocol. The average score captures partial task progress, and the success rate measures binary task completion.

Tables 7 and 8 report the per-dimension average score and success rate, respectively, complementing Table 4. Baseline results are taken from the official RoboDojo simulation leaderboard as of September 2026.

	Generalization						
Method	Std.	Rand.	Precision	Long-Horizon	Memory	Open	Avg.
π_0 (Black et al., 2024)	7.18	0.71	3.56	6.19	3.47	0.25	3.48
$\pi_{0.5}$ (Intelligence et al., 2025)	20.93	5.82	12.40	23.54	5.89	1.98	11.44
X-VLA (Zheng et al., 2026)	17.90	3.04	18.32	16.53	4.76	0.55	10.13
Fast-WAM (Yuan et al., 2026)	4.33	0.34	1.96	9.14	3.55	0.42	3.48
OpenWAM-α (Wang et al., 2026a)	33.16	8.26	18.45	34.93	10.41	1.41	17.18
Xiaomi-Robotics-1 (Guo et al., 2026)	35.65	11.44	26.69	38.39	7.81	3.94	20.07
GPT-6-Astra (Zhang et al., 2026c)	35.32	31.40	12.65	21.45	43.04	34.36	28.97
VPP2 (Ours)	36.39		43.03	56.28	56.01	4.61	**39.26**

Table 7: Per-dimension average score on the RoboDojo simulation benchmark. Generalization is evaluated under standard (Std.) and randomized (Rand.) settings. Avg. is the mean over the five capability dimensions, where the generalization score is the mean of the Std. and Rand. settings. For VPP2, we report the generalization result pooled over both settings (25 episodes each, following the official protocol), which equals their mean. Baseline results are taken from the official leaderboard.

Method	Generalization		Precision	Long-Horizon	Memory	Open	Avg.
	Std.	Rand.					
π_0 (Black et al., 2024)	4.89	0.22	0.75	2.00	2.11	0.25	1.53
$\pi_{0.5}$ (Intelligence et al., 2025)	14.89	1.44	5.50	14.67	4.67	1.67	6.93
X-VLA (Zheng et al., 2026)	12.22	1.33	12.00	9.75	3.56	0.50	6.52
Fast-WAM (Yuan et al., 2026)	2.11	0.11	0.00	5.17	3.44	0.42	2.03
OpenWAM-α (Wang et al., 2026a)	25.56	4.11	9.25	25.33	9.11	1.08	11.92
Xiaomi-Robotics-1 (Guo et al., 2026)	28.00	6.00	18.83	23.67	6.56	3.58	13.93
GPT-6-Astra (Zhang et al., 2026c)	32.67	28.33	4.00	8.25	38.67	31.00	22.48
VPP2 (Ours)	29.50		35.17	37.00	55.56	4.08	**32.26**

Table 8: Per-dimension success rate (%) on the RoboDojo simulation benchmark, computed in the same way as Table 7.